

Università di Verona

Master's Degree in Data Science

COMPUTER VISION

Project report :
*Segmentation and classification on
fundus images*

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Contents

1	Contextualization, task definition and associated application	2
2	State of the art	5
3	Objectives	5
4	Methodology	5
4.1	Data set description	6
4.2	Model	6
4.3	Loss Functions	7
4.3.1	Classification function	7
4.3.2	Segmentation loss	7
4.4	Main functions in the code	8
5	Results	8
5.1	comparison of the results with train_loop_basic and test_loop_basic	8
5.1.1	U-net with ResNet18	9
5.1.2	U-net with ResNet34	11
5.2	With data augmentation	12
6	Conclusion	13

1 Contextualization, task definition and associated application

A fundus examination allows the observation of the structures of the eye located behind the lens, particularly the retina. This medical imaging is then used to examine the blood vessels and the optic nerve and thus allows the detection of possible damage to the retina.

Our goal is to do a classification to determine if an eye, based on its fundus image, has an Age-related Macular Degeneration or a Diabetic Retinopathy or Glaucoma or is normal with no disease.

At the same time we will also do a segmentation on fundus vessels in order to allow some calculus such as the diameter of the vessels which allows the detection of high blood pressure which causes the arteries in the retina to narrow.

In a second time we will complexify the task by doing again the same classification and the segmentation but this time in order to provide a better visualization, particularly when the quality of the fundus is impaired by for example a disease called cataract which opacifies the lens. Indeed, It is difficult to see the vessels on blurred images or with a low contrast so a segmentation could be useful for the detection of vessels when the fundus image doesn't have a good quality due to disease like cataract for example.

Definitions :

Glaucoma : disease that damages optic nerves because of too high eye pressure

Diabetic Retinopathy : damage of blood vessels because of diabetes

Age-related Macular Degeneration : progressive eye disease that causes blurred vision

Cataract : clouding of the eye's natural lens, which normally focuses light onto the retina to provide clear vision

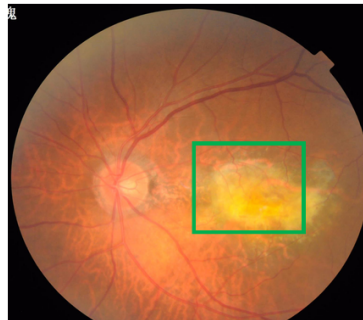


Figure 1: Age-related Macular Degeneration

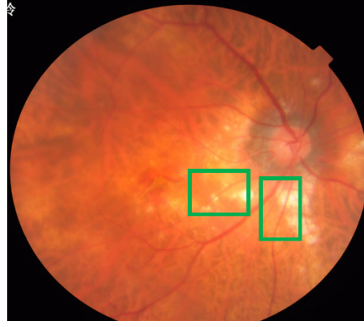


Figure 2: Age-related Macular Degeneration

Age-related Macular Degeneration can be recognized on a fundus image by yellow, white, and round spots

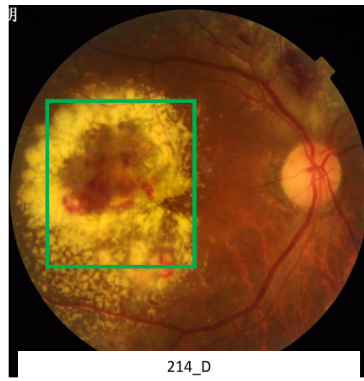


Figure 3: Diabetic Retinopathy

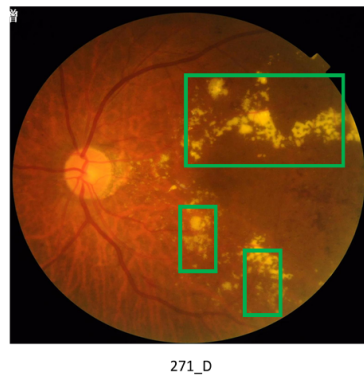


Figure 4: Diabetic Retinopathy

Diabetic Retinopathy, can be recognized by red and yellow spots. The yellow spots are indicative of this disease, but the red spots may be caused by other factors.

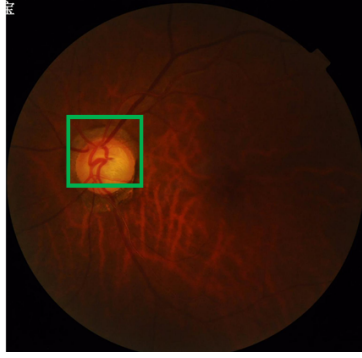


Figure 5: Glaucoma

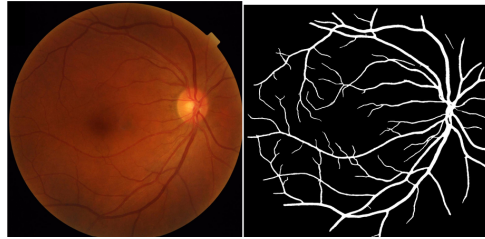


Figure 6: Fundus and the segmentation mask

Glaucoma, can be recognized on a fundus image by white color of the optic nerve and excavation

In our dataset there are not lots of poor quality images that's why we will apply data augmentation by flipping blurred images original y present in our dataset. We can also apply another data augmentation which blurs the images in order to make our model more efficient to segmentate vessels on lower resolution images.

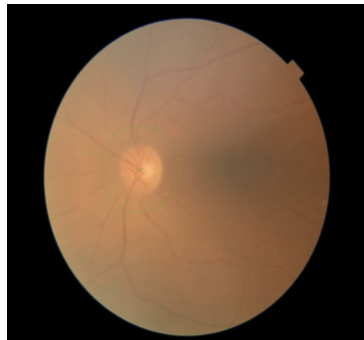


Figure 7: Patient suffering from cataract

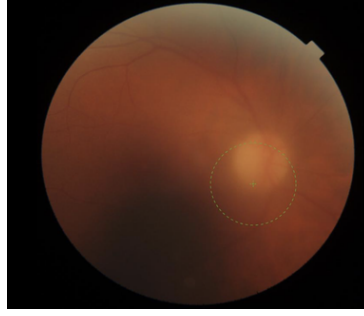


Figure 8: Another patient suffering from cataract

In our dataset there are not lots of poor quality images that's why we will apply data augmentation by flipping blurred images original y present in our dataset. We can also apply another data augmentation which blurres the images in order to make our model more efficient to segmentate vessels on lower resolution images.

2 State of the art

Segmentation of eye fundus is an important task in the medical field. Indeed, it enables automatic disease detection like glaucoma or diabetes. Before the deep learning era, methods used were basically based on increasing contrast and detecting contours but they were too sensitive to luminosity variation, reflections and other factors. Deep learning really improved performance of segmentation detection. In the medical imaging field, the U-net architecture is the most popular architecture for detection and segmentation tasks. This is thanks to its U structure which consists of a contracting path followed by an expanding path. The first one extracts main features of the image and reduces its dimensions whereas the second one restores the original size and creates the segmentation map. Nowadays, there are many versions of U-net that include modern technologies. The main ones are attention U-net, which uses the attention mechanism to focus on the relevant regions, Residual U-net, which uses residual connections to improve learning, and Swin-UNet, where the expanding path and the contracting path are replaced by transformers.

3 Objectives

The objective of our project is to detect blood vessels by segmenting eye fundus images and to classify fundus eye images by predicting the good category among four diseases : age-related macular degeneration, diabetic retinopathy, glaucoma and normal eye. To do so, we tested several models and saw which one had the best results.

4 Methodology

We trained two models based on U-net to simultaneously segment fundus images and classify them as glaucoma, diabetic retinopathy, age-related macular degeneration and normal fundus. To do that, we set a penalty on the segmentation loss and the classification

loss to avoid that one dominates the other. That avoids having a model good at segmentation but not at classification or vice-versa, since the optimizer changes weights based on the total loss.

We wanted then to evaluate our model on images affected by cataract. Since we didn't find a dataset to train our model on this disease, we computed the segmentation loss and the classification loss only for blurred and low-contrast images because they are the closest to fundus images that suffers from cataract. This way, we have an estimation of the performance of our model on these kinds of images. We compared results obtained before and after data augmentation.

4.1 Data set description

The data set that we used contains 600 fundus images and their associated masks for the training and 200 images with the corresponding masks for the test. There is also a file that contains information for each image of the data set : a number (that identifies an image), the disease associated with the image fundus and a binary score for low contrast, blur and color illumination (0 refers to poor quality and 1 to good quality). The data set is available here : <https://www.kaggle.com/datasets/nikitamanaenkov/fundus-image-dataset-for-vessel-segmentation/data>.

	train	test
Number of images	600	200
Age-related Macular Degeneration	25 %	25 %
Diabetic Retinopathy	25 %	25 %
Glaucoma	25 %	25 %
no diagnosed eye disease	25 %	25 %
blurred image	15 %	20,5 %
Low contrast image	3,33 %	7,5 %

Figure 9: Data set description

4.2 Model

U-net architecture

This model is based on the U-net architecture. That means that it is composed of a contracting path, an expanding path, a bottleneck and skip connections. In the contracting path there are convolutions and poolings that reduce spatial dimensionality of the feature map and increase its depth. This way, the model can extract features from the input data. In the expanding path, image size is increased by using convolutional layers, upsampling and combining information from the contracting path using skip connections. Skip connections help the expanding path get details that have been lost when shrinking the image. The bottleneck is where the most compressed information is stored. It links the expanding path with the contracting path.

ResNet

In our work, we used a U-net architecture where the contracting path is replaced by a ResNet architecture. That enables the model to make classification on different diseases

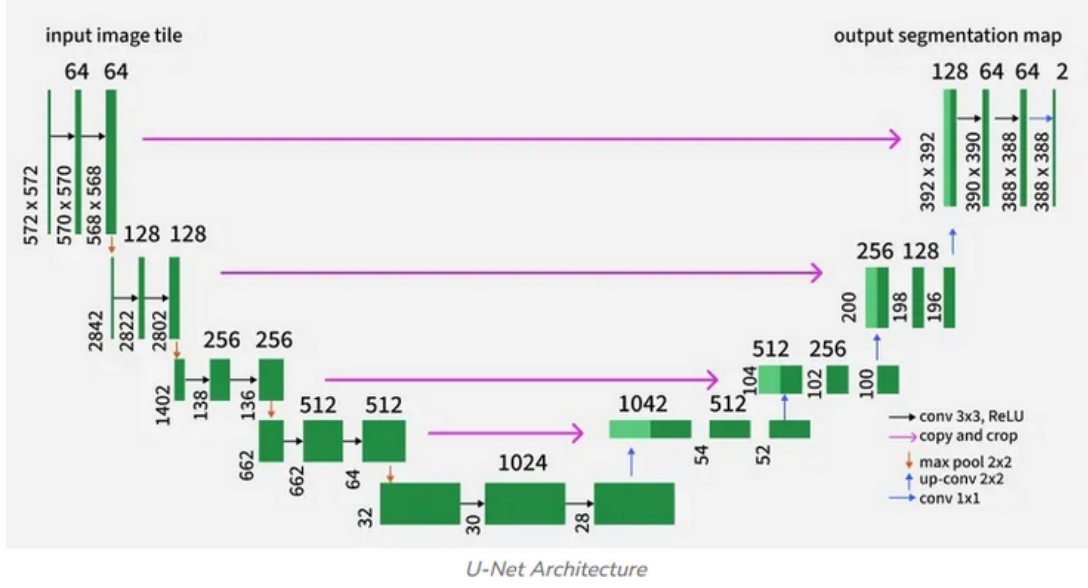


Figure 10: U-net structure

in an efficient way. We used ResNet34 and ResNet18 for our experiments, depending on the complexity of the dataset.

4.3 Loss Functions

4.3.1 Classification function

As we want to do a multiclass classification, we used the CrossEntropyLoss function. For each sample in the dataset, the cross-entropy loss quantify how well the model's prediction matches the true label. A lower loss for a sample indicates a more accurate prediction.

$$\ell(x, y) = L = \{l_1, \dots, l_N\}^\top, \quad l_n = -w_{y_n} \log \frac{\exp(x_{n, y_n})}{\sum_{c=1}^C \exp(x_{n, c})} \cdot 1\{y_n \neq \text{ignore_index}\}$$

Figure 11: Cross entropy formula

x_{n, y_n} represents the logit predicted by the model, y represents the target class and $x_{n, c}$ represents the logits of the good class.

4.3.2 Segmentation loss

We used the Binary Cross Entropy With Logits Loss as a segmentation loss function.

$$\ell(x, y) = L = \{l_1, \dots, l_N\}^\top, \quad l_n = -w_n [y_n \cdot \log x_n + (1 - y_n) \cdot \log(1 - x_n)]$$

Figure 12: BCE formula

We chose the Binary Cross Entropy With Logits Loss instead of the Binary Cross Entropy because the first one because the output of our model is raw scores so computation is more stable with the Binary Cross Entropy With Logits Loss.

4.4 Main functions in the code

The main functions of our code are:

`train_loop_basic` : it enables to run the train by applying a penalty on the segmentation loss and the classification loss on the original dataset that is to say before the data augmentation.

`train_loop` : it enables to run the train by applying a penalty on the segmentation loss, on the classification loss and also on the blurred images and those with low contrast, which can be useful to detect cataract since the fundus image of an eye with cataract presents these same characteristics.

`train_loop_faster` : it is a faster version of `train_loop`. It avoids applying the loss function on each blurred image or image with low contrast by using tensors.

`test_loop_basic` : it enables run the model on the test set using the results of `train_loop_basic`.

`test_loop` : it enables run the model on the test set.

`test_loop_faster` : it is a faster version of `test_loop` that. It avoids applying the loss function on each blurred image or image with low contrast by using tensors.

`get_images` : it retrieves the path of high-quality images and the path of the associated mask.

`data_augmentation` : it applies data augmentation by flipping images along the vertical axis (to have left eyes and right eyes).

5 Results

5.1 comparison of the results with `train_loop_basic` and `test_loop_basic`

	ResNet18	ResNet34
Classification Loss	0.020	0.026
Segmentation Loss	0.005	0.003
Total Loss	0.025	0.029
Accuracy A	74%	44%
Accuracy D	94%	64%
Accuracy G	74%	96%
Accuracy N	52%	58%

Figure 13: Results with `test_basic`

The accuracies of the 4 diseases are better with the ResNet18 than with the ResNet34. In fact without data augmentation our dataset has 1000 images so we were thinking about overfitting. By taking a model with less parameters we limit this overfitting.

5.1.1 U-net with ResNet18

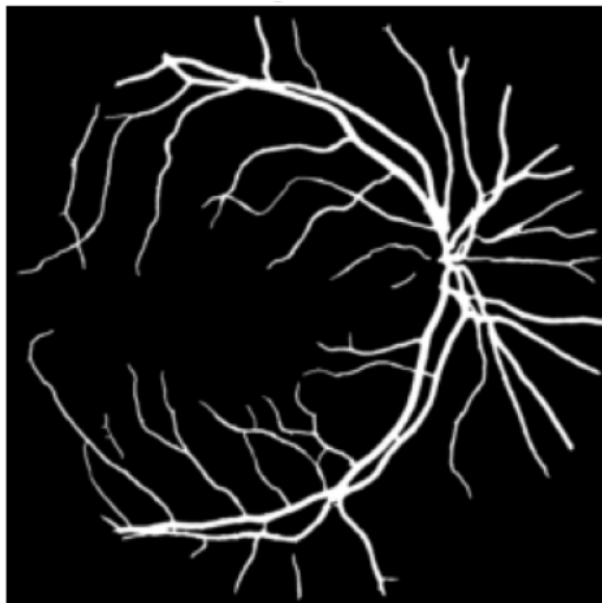


Figure 14: Segmentation mask (ground truth) at epoch 10 of the train (ResNet18)

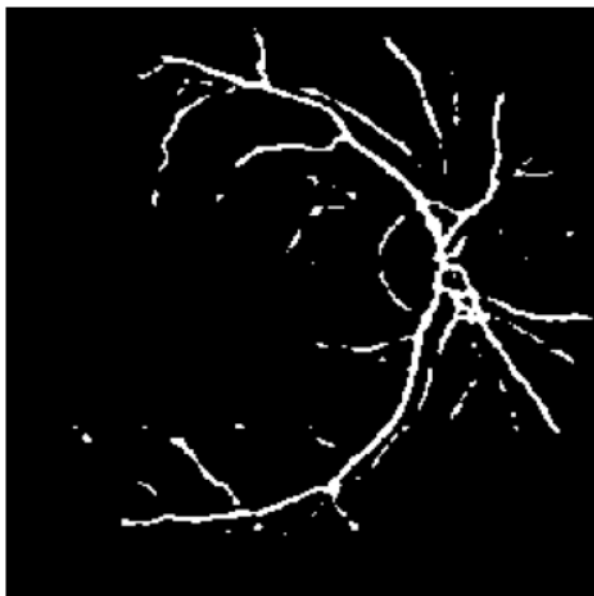


Figure 15: Predicted mask at epoch 10 of the train (ResNet18)

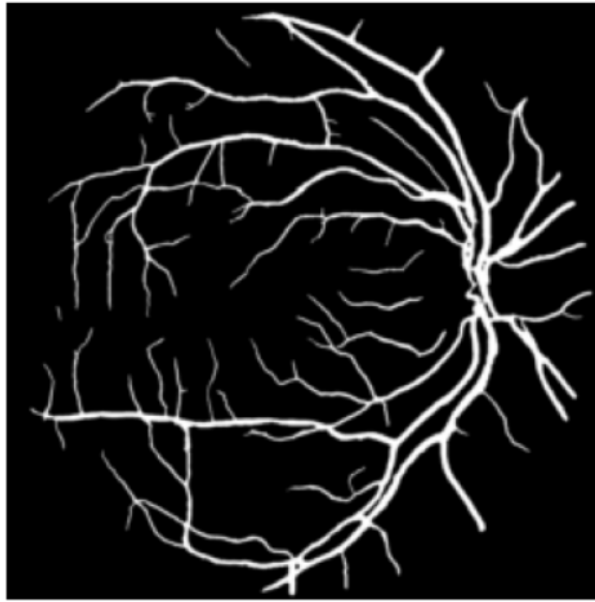


Figure 16: Segmentation mask (ground truth) during test (ResNet18)

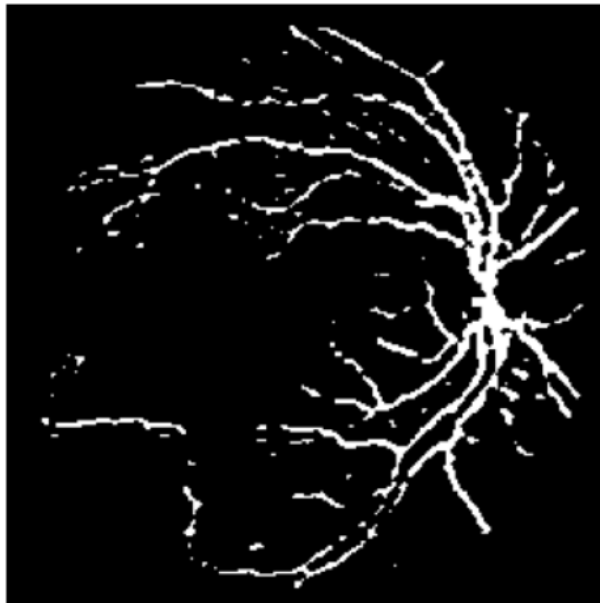


Figure 17: Predicted mask during test (ResNet18)

5.1.2 U-net with ResNet34

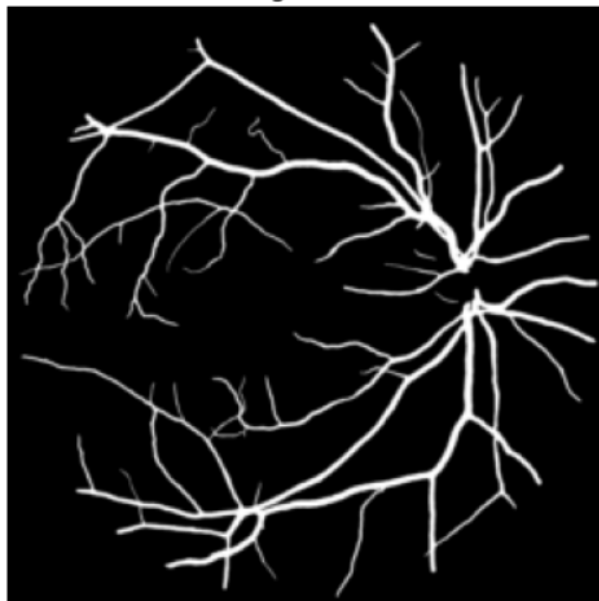


Figure 18: Segmentation mask (ground truth) during test (ResNet34)

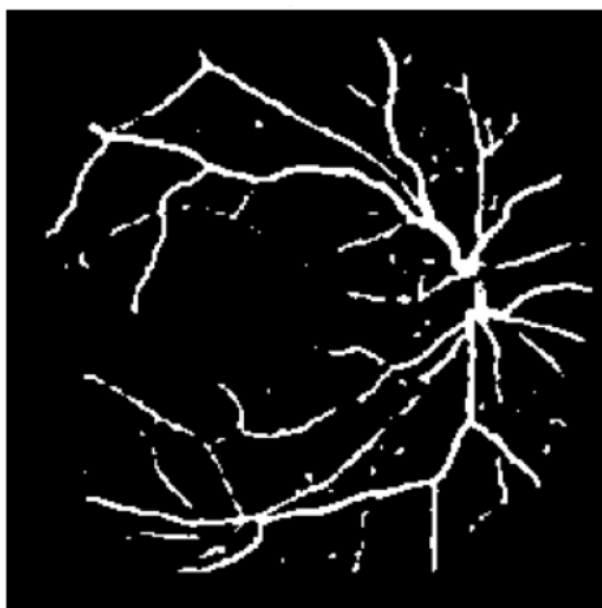


Figure 19: Predicted mask during test (ResNet34)

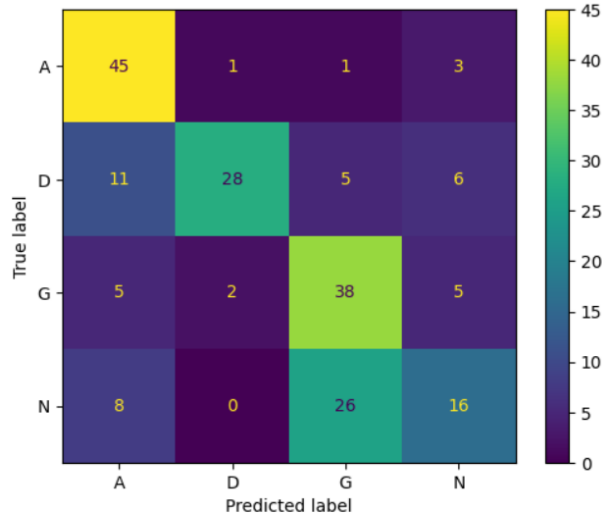


Figure 20: Confusion matrix (ResNet34)

We can notice from the table shown in figure 5 that U-net combined with ResNet18 has better results for classification and segmentation. This is confirmed by the confusion matrix on figure 6 and the predicted mask segmentation at the last epoch of training. Indeed, the one corresponding to ResNet18 is more precise and similar to the ground truth than the one corresponding to ResNet34.

5.2 With data augmentation

To do data augmentation, we selected 20% of high-quality images and we flipped them. We chose as percentage 20% in order to not have too many high-quality images in the dataset after data augmentation compared to before data augmentation. We also blurred 20% of the good images. However, the results were not conclusive.

	ResNet34	ResNet18
Classification Loss	1.96	2.28
Segmentation Loss	1.05	0.99
Accuracy A	24%	12%
Accuracy D	52%	0%
Accuracy G	30%	56%
Accuracy N	6%	16%

Figure 21: Comparison of the results with $\alpha = \beta = 1$

	ResNet34
Classification Loss	2.01
Segmentation Loss	1.09
Accuracy A	10%
Accuracy D	64%
Accuracy G	12%
Accuracy N	0%

Figure 22: Comparison of the results with $\alpha = 1$ and $\beta = 1.5$

6 Conclusion

In order to segment and classify fundus images simultaneously, we used a U-net model where the encoder was replaced by a ResNet model. We used ResNet18 and ResNet34 and compared the results. Then we did a data augmentation to see if this our result can be better. In the future, it can be possible to try other values for the penalty applied for the segmentation loss and the one applied for the classification loss to see if it can improve the results.